

What is ergodicity?

1 Introduction: the concept of ergodicity

Does observing a single instance of a system for a long enough time allow one to understand the statistics of a population of such systems? We say a system is ergodic when the answer to this question is yes. Ergodicity justifies generalising temporal observations of an individual experiment, agent or organism to population-level statistical statements. It also justifies specialising population level statistical properties to statements about individual members of a population. The scientific study of ergodicity grew out of physics, where it provided justification for the use of ensembles in statistical mechanics, and out of statistics where it provided justification for the efficacy of statistical inference. The reason researchers needed a word for this property is that it was quickly understood by physicists, statisticians and mathematicians that it does not always hold and should therefore not be treated as an unexamined assumption. In this volume, we have assembled a diverse collection of topical research papers that have in common some variation of the “ergodicity question”: is it legitimate to replace statistics gathered from an individual over time with statistics over a population or an ensemble? Since the answer to this question is often no, much of the interesting science comes from examining how to proceed when the ergodicity assumption fails. In addition to some contributions from physics [13, 17, 22, 29], interdisciplinary topics include contributions from behavioural science [19, 23, 30], reinforcement learning [6, 31], economics [2, 18], climate change [24], psychology [11], cognitive science [12] and time series analysis [9].

Despite the fundamental, and fairly intuitive, nature of the concept of ergodicity, it turns out to be difficult to make it mathematically precise in a way that applies to all use cases. As a consequence, a large distance has opened up between fundamental research in the mathematical sciences and applications of ergodicity in the physical, social and economic sciences. Indeed, it can sometimes seem that fundamental notions of ergodicity from dynamical systems theory have very little to do with practical instances of the ergodicity question. In this introduction, we provide a brief tour through the main mathematical ideas about ergodicity and try to link them to examples of the ergodicity question appearing in the applications mentioned above. Our writing is necessarily high-level, and we intentionally omit many of the mathematical qualifications necessary to make our statements mathematically precise. The objective is to convey ideas, with the help of illustrative examples. For readers wishing to delve into the mathematical details, there are several excellent treatises available, e.g. [4, 20, 25, 32].

Before diving in, let us outline the structure of this in-

roduction and the rationale linking the sections. We begin with a description of ergodicity from the perspective of deterministic dynamical systems in Section 2. We describe how the study of recurrences and the existence of time averages leads to the idea of invariant sets, invariant measures, and an abstract definition of ergodicity in terms of the metric indecomposability property: all invariant sets have measure zero or one. We try to communicate the fact that invariant measures are usually not unique and that the choice made in deciding which invariant measure to study has practical implications for what aspects of a dynamical system are typical. In Section 3 we describe how stationary stochastic processes can be incorporated into the framework of Section 2. The unification of the study of recurrences in deterministic and random dynamics into a single conceptual framework is one of the great intellectual achievements of ergodic theory. Up to this point, we have considered only systems in which the total measure is finite. In Section 4 we consider what happens when this requirement is relaxed and introduce the topic of infinite ergodicity. Infinite ergodic theory describes the time asymptotics of cumulative sums for recurrent processes that are not stationary in the usual sense since the expected return time is infinite. Several new phenomena appear, including non-trivial scaling of cumulative sums, random limits and loss of self-averaging. In Section 5 we consider genuinely non-stationary stochastic processes and describe how appropriate transformations can sometimes be found to construct an ergodic process. This is important for many applications and allows one to reason about differences between individual trajectories and ensemble properties in non-stationary problems where neither the usual time nor ensemble averages exist. Finally in Section 6 we consider more operational notions of ergodicity which can characterise the ergodicity question in practical problems where both the sample size and the time observation window are finite. Such operational notions are necessarily non-binary in nature and must allow for uncertainty, which can be characterised by statistical testing. They have the strong advantage of being applicable to a wider range of use cases than more mathematically strict notions.

2 Mathematical formulation of ergodicity for deterministic dynamical systems

We consider a deterministic dynamical system with discrete time ($n \in \mathbb{N}$) of the form

$$x_{n+1} = T(x_n), \quad (1)$$

where $x_n \in X$ describes the state of the system at time n , and the map $T : X \rightarrow X$ denotes the time-one evolution. Moreover, we write $x_{n+m} = T^m(x_n)$ for any $m \in \mathbb{N}$.¹

¹In the current exposition we restrict the discussion to discrete time, but continuous time generalisations are straightforward.

It is natural to address the question whether for an observable $f : X \rightarrow \mathbb{R}$ the time-average of this observable along an orbit $\{x_i\}_{i \in \mathbb{N}}$

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n(x_0)) \quad (2)$$

exists. If so, this knowledge can be useful for the characterisation and understanding of the dynamics of the system. It turns out that in general this question is not easy to answer, but ergodic theory provides important insights.

To make things less abstract, we consider a specific (classical) example, the doubling map on $X = [0, 1]$ given by

$$T(x) = 2x \mod 1 = \begin{cases} 2x & \text{if } 0 \leq x \leq \frac{1}{2} \\ 2x - 1 & \text{if } \frac{1}{2} < x < 1. \end{cases} \quad (3)$$

See, for example, [25, chap3]. A remarkable result from ergodic theory asserts that the average (2) exists (if f is integrable) and is equal to $\int_0^1 f(x)dx$ for a set of initial conditions $x_0 \in X_0 \subset [0, 1]$ of full Lebesgue measure.

The chosen example has the property that when writing $x := 0.b_1b_2b_3b_4\dots$ in binary form with $x = \lim_{N \rightarrow \infty} \sum_{j=1}^N b_j 2^{-j}$ and $b_j \in \{0, 1\}$, we have $T(x) = 0.b_2b_3b_4\dots$. So T effectively induces a left shift on the binary digits. Moreover, the binary digits have geometric meaning: $x \in [0, \frac{1}{2}]$ if $b_1 = 0$ and $x \in [\frac{1}{2}, 1]$ if $b_1 = 1$, so that $\chi_{[0, \frac{1}{2}]}(T^m(x)) = 1$ precisely when $b_{m+1} = 0$.

That the average (2) does not always exist is easily illustrated by choosing the observable $f = \chi_{[0, \frac{1}{2}]}$, the characteristic function

$$\chi_{[0, \frac{1}{2}]}(x) = \begin{cases} 1 & \text{if } x \in [0, \frac{1}{2}], \\ 0 & \text{if } x \notin [0, \frac{1}{2}]. \end{cases}$$

For instance, if $x_0 = 0.011000011111111\dots$ (continuing the binary expansion by doubling the number of equal consecutive digits), then the partial average $s_N := \frac{1}{N} \sum_{n=0}^{N-1} f(T^n(x_0))$ keeps oscillating between $\frac{1}{3}$ and $\frac{2}{3}$ as $N \rightarrow \infty$ and so does not converge. Indeed, as this construction relies on the tail of the binary digits as $N \rightarrow \infty$, it follows that the time-average does not exist for a dense set of initial conditions: i.e. every interval, however small, contains an initial condition for which the time-average does not exist.

Moreover, even if the time-average exists, it may well converge to a value not equal to $\int_0^1 \chi_{[0, \frac{1}{2}]}(x)dx = \frac{1}{2}$. Namely, for every x_0 whose binary expansion has a periodic tail, repeating a block with p zeros and q ones, the time-average exists and is equal to $p/(p+q)$. It follows that the set of initial conditions for which the time-average exists but is not equal to $\frac{1}{2}$ is also dense.

This example illustrates that the fact that the time-average is equal to $\frac{1}{2}$ for all but a zero Lebesgue measure set of initial conditions is far from obvious, and relies on ignoring a topologically large set of initial conditions. Indeed, the perceived importance of this fact depends on how much we care about the Lebesgue measure (volume on $[0, 1]$). We may for instance assert that a typical initial condition $x_0 \in [0, 1]$ should lie in the "good" full Lebesgue measure set of initial conditions $\hat{X}$. However, if we simulate the system on a computer on which all numbers have a finite binary representation, then for all initial conditions on the computer we find $\lim_{n \rightarrow \infty} T^n(x_0) = 0$ so that the time-average (2) converges to $f(0)$ (and indeed no initial condition on the computer lies in the good set $\hat{X}$).

In ergodic theory, the existence and persistence of time-averages is associated with so-called ergodic invariant probability measures. One considers a measure-preserving dynamical system (X, μ, T) where

- X is the phase space - the set of all possible states the system can be in,
- μ is a measure which assigns a non-negative volume to any subset of the phase space,
- $T : X \mapsto X$ is a map that generates dynamics on X by iteration (1),

and measure-preserving means that

$$T_*\mu(A) := \mu(T^{-1}A) = \mu(A) \quad (4)$$

for all (μ -measurable) subsets $A \subset X$, where $T^{-1}(A)$ denotes the pre-image of A under T :

$$T^{-1}(A) = \{x \in X : T(x) \in A\}. \quad (5)$$

In other words, as $T^{-1}(A)$ is the set of all points in X that are mapped to A by T , the map T induces a redistribution of the measure μ (without shrinking or increasing it). A set $A \subset X$ is said to be *invariant* if $T^{-1}(A) = A$.

We can now state the mathematical definition of ergodicity: a measure preserving dynamical system is ergodic if all invariant sets A have the property that $\mu(A) = 0$ or $\mu(X \setminus A) = 0$. If μ is a probability measure (i.e. $\mu(X) = 1$), then the more commonly stated (but equivalent) condition for ergodicity is that $\mu(A) \in \{0, 1\}$.

Birkhoff's Ergodic Theorem [7] asserts that for an ergodic probability measure-preserving dynamical system (X, μ, T)

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n(x_0)) = \int_X f(x) d\mu(x) =: \mathbb{E}_\mu[f] \quad (6)$$

for all integrable functions f on X and μ -almost all initial conditions x_0 . It is important to note that this theorem not only establishes the existence of the time-average, but also that it is essentially constant on the supports of ergodic invariant measures (which are called ergodic components).

Boltzmann's Ergodic Hypothesis in (statistical) physics asserts that the Liouville (canonical phase volume) measure is ergodic, in which case (6) is usually summarized informally as that the time-average of an observable along orbits is typically equal to the (phase) space average of the observable f .

To reflect on our example of the doubling map, above, we find that the different observed time-averages relate to different ergodic probability measures: the Lebesgue measure and equally distributed atomic measures on periodic orbits (which are dense).

We note that dynamical systems on a compact state space always have at least one invariant probability measure. Moreover, the set of invariant probability measures of a dynamical system forms a convex set (since when μ_1 and μ_2 are invariant probability measures, then so is $s\mu_1 + (1-s)\mu_2$ for any $s \in [0, 1]$). It turns out that the extremal points of the set of invariant probability measures (i.e. the ones that cannot be written as a weighted average of two others) are precisely the ergodic ones. Vice versa, every non-ergodic invariant measure is the weighted average of two or more ergodic invariant measures. For more details of this kind, cf. [32].

If a dynamical system has many ergodic invariant measures (which is not unusual), the specific statistic of a given trajectory can be viewed to depend on the initial condition. Indeed, distinct ergodic invariant probability measures are mutually singular (they are concentrated on disjoint measurable sets of full measure), so that almost every initial condition is associated with at most one statistic. There exist variations of Birkhoff's Ergodic Theorem (that we do not state here in detail) which express this dependence by the introduction of a conditional expectation in the right-hand side of (6) [32].

It is important to note that ergodic invariant measures essentially represent information about recurrence: for μ -almost all initial conditions, the proportion of points in its trajectory that lie in A is equal to $\mu(A)$. Moreover, with the first hitting time to A for a point $x \in X$ defined as $\tau_A(x) := \min\{n \geq 1 : T^n(x) \in A\}$, the μ -expectation of the first return time from A to itself is $\mathbb{E}_\mu(\tau_A | A) := \mu(A)^{-1} \int_A \tau_A(x) d\mu(x) = \mu(A)^{-1}$ (Kac's Lemma [15]).

The study of ergodicity in dynamical systems was originally motivated to a large extent by Boltzmann, whose work on statistical mechanics gave rise to the hypothesis that the (naturally invariant) Liouville measure in classical mechanics is ergodic. It took many decades of work, with major contributions by Sinai and co-workers, to prove that this hypothesis holds for a many particle hard ball gas [26]. But as a counterpoint, it was also shown that for soft ball gases the hypothesis typically does not hold [28]. Indeed, as argued by Lebowitz [16], Goldstein [14] and others, ergodicity of the Liouville measure may be viewed as an effective paradigm and

powerful approximation in appropriate settings, rather than a strict requirement for statistical mechanics to work. Even for some integrable systems like the Toda model [17] it has been shown that important features of equilibrium statistical mechanics hold, despite the fact that such systems cannot exhibit the molecular chaos associated with the ergodic hypothesis. Many physicists now believe that what really matters is that, for a broad class of stationary many-body physical systems, most/many trajectories correspond to thermodynamic equilibrium states. This viewpoint is sometimes referred to as the typicality account of ergodicity (see also [22]).

3 Stationary stochastic processes as dynamical systems

It may not be immediately clear how the above notion of ergodicity developed for deterministic dynamical systems applies to evolutions that contain stationary stochastic processes. Here, by stationarity, we mean that the joint law of the process does not change under time translation.

Stochastic processes relate to a random form of dynamics, for example encoded by the transition kernel of a Markov chain or by the explicit presence of a stochastic driving term in an otherwise deterministic evolution, such as appears in stochastic differential equations.

The definition of ergodicity in terms of invariant measures as given in the previous section at first sight does not appear to be appropriate. However, there is a way to view a stationary stochastic process as arising from an ergodic deterministic shift dynamical system on its path space.

3.1 Canonical (path space) representation of a stationary stochastic process

In statistical applications, one typically begins with a discrete-time stochastic process $\{X_n\}_{n \in \mathbb{Z}}$ taking values in a state space S . Stationarity implies that the marginal distribution of X_n at any single time n is given by a time-invariant probability measure π on S , and more generally, that all finite-dimensional joint laws are invariant under time translation.

To view this statistical process as a deterministic dynamical system, we lift the evolution to the *path space* of bi-infinite trajectories,

$$\Omega = S^{\mathbb{Z}}. \quad (7)$$

An individual sample path is a sequence $\omega = (\dots, \omega_{-2}, \omega_{-1}, \omega_0, \omega_1, \omega_2, \dots) \in \Omega$, where $\omega_n = X_n(\omega)$ represents the state at time n . The stochastic evolution over time is then encoded by the deterministic (left) *shift map* $\theta : \Omega \rightarrow \Omega$, defined as

$$(\theta \omega)_n = \omega_{n+1}, \quad \forall n \in \mathbb{Z}. \quad (8)$$

The probabilistic structure of the process enters this dynamical picture via the joint laws: by the Kolmogorov extension

theorem [8], the collection of stationary joint distributions uniquely induces a probability measure μ on the path space Ω . Stationarity of the process directly translates to the shift-invariance of μ , i.e., $\mu(\theta^{-1}A) = \mu(A)$ for all measurable sets of paths $A \subset \Omega$.

When stochastic processes are used in applications or modelling, we do not usually compute time and ensemble averages over full trajectories in path space. Typically, one considers observables f that depend only on the current state, represented by $f(\omega) = \hat{f}(\omega_0)$ for some function $\hat{f} : S \rightarrow \mathbb{R}$. Assuming the path-space measure μ is ergodic under the shift map θ , Birkhoff's Ergodic Theorem yields for μ -almost all $\omega \in \Omega$ that

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \hat{f}(X_n(\omega)) &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(\theta^n \omega) \\ &= \int_{\Omega} f(\omega) d\mu(\omega) \\ &= \int_S \hat{f}(x) d\pi(x) \\ &= \mathbb{E}_{\pi}[\hat{f}] \end{aligned} \quad (9)$$

for μ -almost every path $\omega \in \Omega$. This is the standard equality expressing ergodicity for stationary stochastic processes.

While the canonical representation is general for any stationary process, the induced measure μ is potentially very complicated since it encodes all dynamical structure, such as temporal correlations and finite-dimensional marginal distributions. The simplest arises when the process is i.i.d. and the path space measure is the product measure $\mu = \pi^{\mathbb{Z}}$.

3.2 Random dynamical systems

Random dynamical systems are dynamical systems that are driven by a stochastic process.

$$x_{n+1} = F(x_n, \omega_n), \quad (10)$$

where the x_n take values in a state space, X , and the ω_n are random variables taking values in a "noise" space S . The noise is assumed to be generated by an ergodic stationary stochastic process with stationary measure μ , like discussed in Section 3.1. We can thus view the evolution of the noise as ergodic shift dynamics on the path space of the noise $\Omega = S^{\mathbb{Z}}$, and subsequently consider the random dynamical system (10) as a dynamical system on $X \times \Omega$, generated by the time-one evolution map

$$(x, \omega) \rightarrow \Theta(x, \omega) := (F(x, \omega_0), \theta\omega). \quad (11)$$

The skew-product structure is characteristic of driven systems, where the state-space dynamics on X depend on the noise process on Ω , but not vice versa. This formulation provides a natural framework for physical systems driven by exogenous environmental fluctuations. In continuous time,

stochastic and random differential equations are naturally viewed in this setting [3], avoiding the semimartingale restrictions required by Itô calculus.

Furthermore, under mild assumptions, every Markov process $\{x_n\}_{n \in \mathbb{Z}}$ on $X \subseteq \mathbb{R}$ (as well as finite-memory recursive models like ARMA processes) admits a skew-product representation:

$$x_{n+1} = F(x_n, u_n), \quad (12)$$

where $(u_n)_{n \in \mathbb{Z}}$ are i.i.d. uniform random variables on $[0, 1]$. This representation—often called the *random mapping* or *functional representation* of the Markov chain—follows directly from inverse transform sampling. For a transition kernel $P(x, \cdot)$ with conditional cumulative distribution function $\mathcal{F}_x(y) := P(x, (-\infty, y])$, setting

$$F(x, u) = \mathcal{F}_x^{-1}(u) := \inf\{y \in \mathbb{R} : \mathcal{F}_x(y) \geq u\} \quad (13)$$

recovers the transition kernel $P(x, \cdot)$ when $u \sim \text{Unif}(0, 1)$.

Let us illustrate the skew-product construction for the elementary example of the discrete Ornstein-Uhlenbeck process:

$$x_{n+1} = a x_n + \omega_n, \quad (14)$$

where $|a| < 1$ and $\omega_n \in S$ are i.i.d. random variables with probability measure π on S . The interpretation of the path $\omega = (\dots \omega_{-2}, \omega_{-1}, \omega_0, \omega_1, \omega_2 \dots) \in \Omega$ is that of a noise realisation. The i.i.d. noise is represented by the ergodic dynamical system $(\Omega, \pi^{\mathbb{Z}}, \theta)$, with $\Omega := S^{\mathbb{Z}}$.

In the random dynamical systems setting, when contemplating the existence of ergodic invariant measures of the (deterministic) skew-product dynamical system, it is important to keep in mind that the probabilistic assumptions on the noise are represented by a particular (choice of the) ergodic invariant measure μ . In practice, this means that we are interested in ergodic invariant measures of the skew-product system for which the marginal on Ω is equal to μ . Ergodic invariant measures with different Ω -marginals de facto relate to random dynamical systems with different noise models.

We now proceed to show how the stationary measure of a random dynamical system relates to an ergodic invariant measure of the skew product representation with the appropriate Ω -marginal. Let ν denote an ergodic invariant measure on $X \times \Omega$ with Ω -marginal $\mu(U) := \nu(X \times U)$ for any (measurable) $U \subset \Omega$. Furthermore, the measure π on X is defined as the X -marginal of ν , i.e. $\pi(A) := \nu(A \times \Omega)$.

Then for π -integrable observable $f : X \rightarrow \mathbb{R}$, with $\hat{f}(x, \omega) := f(x)$, Birkhoff's Ergodic Theorem implies that for μ -almost every noise realization ω , convergence of the time average to the ensemble average (with respect to the

stationary measure) holds for ν_ω -almost every x_0 , yielding

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(x_n) &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \hat{f}(\Theta^n(x_0, \omega)) \\ &= \int_{X \times \Omega} \hat{f}(x, \omega) d\nu(x, \omega) \quad (15) \\ &=: \int_X f(x) d\pi(x) =: \mathbb{E}_\pi[f]. \end{aligned} \quad (16)$$

We have used here that ν -almost all (x_0, ω) implies π -almost all $x_0 \in X$ and μ -almost all $\omega \in \Omega$.

The random dynamical systems perspective allows for addressing questions that do not naturally arise from viewing (10) from a stochastic processes perspective. The deterministic skew-product representation allows for asking typical dynamical systems questions, beyond statistics and ergodic theory, about the existence of invariant sets, attractors, bifurcations, even chaos (sensitive dependence on initial conditions). Indeed, random dynamical systems theory pursues dynamical phenomena in the state space X . While it is in general challenging (or impossible) to identify results that hold for all noise realisations, results from ergodic theory can be employed to obtain insights that hold for μ -almost all noise realisations (without identifying precisely the exceptional set of noise realisations).

For the benefit of readers who find explicit examples useful, we continue to discuss a bit more the example of the discrete Ornstein-Uhlenbeck process (14). If we assume noise to be Gaussian, $\omega_n \sim N(m_\Omega, \sigma_\Omega^2)$, then since linear combinations of Gaussians are Gaussian, it follows that the stationary measure π_a of the discrete Ornstein-Uhlenbeck process is also Gaussian:

$$x_n \sim N(m_x, \sigma_x^2), \quad (17)$$

where the mean, m_x , and variance, σ_x^2 , are to be determined. Since $ax_n \sim N(am_x, a^2\sigma_x^2)$ and the fact that at stationarity, $x_{n+1} \stackrel{d}{=} x_n$, a direct calculation yields that

$$m_x = \frac{m_\Omega}{1-a} \quad (18)$$

$$\sigma_x^2 = \frac{\sigma_\Omega^2}{1-a^2}, \quad (19)$$

and the resulting stationary measure π_a of (14) has Lebesgue density

$$\frac{d\pi_a}{dx}(x) = \sqrt{\frac{1-a^2}{2\pi\sigma_\Omega^2}} \exp \left[-\frac{1-a^2}{2\sigma_\Omega^2} \left(x - \frac{m_\Omega}{1-a} \right)^2 \right]. \quad (20)$$

We may verify Birkhoff's Ergodic Theorem by explicit calculation. For simplicity, let $m_\Omega = 0$. By squaring (14) and rearrangement one obtains:

$$(1-a^2)x_n^2 = \omega_n^2 + 2ax_n\omega_n - (x_{n+1}^2 - x_n^2) \quad (21)$$

Summing from $n = 0$ to $N-1$ and dividing by N yields the time average on the left-hand side:

$$(1-a^2) \frac{1}{N} \sum_{n=0}^{N-1} x_n^2 = \frac{1}{N} \sum_{n=0}^{N-1} \omega_n^2 + 2a \frac{1}{N} \sum_{n=0}^{N-1} x_n \omega_n - \frac{x_N^2 - x_0^2}{N} \quad (22)$$

In the limit $N \rightarrow \infty$, the first term on the right-hand side converges to σ_Ω^2 by the Strong Law of Large Numbers. The second term converges to zero by the Martingale Strong Law of Large Numbers. The third term also converges to zero. Intuitively this is because x_N stays of order one almost surely but a full proof is technical. We thus obtain that the time average of x^2 along a trajectory is equal to the variance σ_x^2 of the stationary distribution of x as expected by ergodic theory, for almost every noise realisation and almost every initial condition x_0 :

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} x_n^2 = \frac{\sigma_\Omega^2}{1-a^2} = \mathbb{E}_{\pi_a}[x^2]. \quad (23)$$

It is interesting to note that (22) in fact holds for all initial conditions x_0 (and almost all noise realisations). This is due to the contractive nature of (14) which renders the random dynamical system *synchronising* in the sense that $\lim_{n \rightarrow \infty} |x_n - \tilde{x}_n| = 0$ for all samples $\{x_n\}_{n \in \mathbb{N}}$ and $\{\tilde{x}_n\}_{n \in \mathbb{N}}$, i.e. all trajectories converge to a so-called *random fixed point*, whose evolution depends on the noise realisation [3].

Gaussian examples tend to be overemphasised in the literature due to their analytical tractability; an interesting non-Gaussian case is symmetric discrete Bernoulli noise

$$\omega_n = \begin{cases} 1 & \text{with probability } \frac{1}{2}, \\ -1 & \text{with probability } \frac{1}{2}. \end{cases} \quad (24)$$

The discrete Ornstein-Uhlenbeck process (14) with this discrete noise is an example of a finitely generated contractive Iterated Function Systems [5]. By considering the extremal noise trajectories ($\omega_n = 1$ and $\omega_n = -1$ for all n) it is easy to show by direct calculation that the stationary measure $\pi_a(x)$ of this process must be compactly supported on $\left[-\frac{1}{1-a}, \frac{1}{1-a}\right]$. Following the reasoning of (21), the time average of x^2 can again be calculated, using the fact that $\omega_n^2 = 1$ and x_n^2 is explicitly bounded by $\frac{1}{(1-a)^2}$. We obtain, for almost all noise realisations,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} x_n^2 = \frac{1}{1-a^2}. \quad (25)$$

Ensemble averages for the Bernoulli case are difficult to calculate explicitly because the nature of the stationary measure π_a , as a function of a , is surprisingly complicated. It has been characterised in [27] as follows:

- $0 < a < \frac{1}{2}$: π_a is singular and supported on a Cantor set of dimension $\log 2 / \log a^{-1}$.
- $a = \frac{1}{2}$: π_a is the uniform Lebesgue measure on $[-2, 2]$
- $\frac{1}{2} < a < 1$: π_a is absolutely continuous with respect to the Lebesgue measure except for a set of values of a of measure zero where it is again singular.

It follows from general properties of contractive Iterated Functions Systems that the stationary measure π_a is unique and the corresponding induced measure on path space is ergodic, even though the measure itself may be complicated. The uniform case, $a = \frac{1}{2}$ is the only case for which explicit calculations are simple, yielding uniform stationary measure and

$$\mathbb{E}_{\pi_{\frac{1}{2}}} [x^2] = \int_{-2}^2 \frac{1}{4} x^2 dx = \frac{4}{3}. \quad (26)$$

This agrees with the time-averaged variance given by (24) with $a = \frac{1}{2}$.

4 Infinite ergodicity

Infinite ergodic theory extends classical ergodic theory to recurrent dynamics possessing an invariant measure that is not normalizable ($\mu(\Omega) = \infty$). A characteristic feature is that any set of positive finite measure, $A \subset \Omega$ with $0 < \mu(A) < \infty$, is visited infinitely often μ -almost everywhere, yet the expected return time $\mathbb{E}[\tau_A]$ is infinite. Mathematically, this occurs when the return-time distribution has heavy tails; physically, it implies that while trajectories revisit A infinitely many times, they spend a vanishing fraction of their total time there because excursions between visits can be arbitrarily long.

Equality of Averages is often trivial because time averages and recurrence frequencies tend to zero. Infinite ergodic theory then has less to do with equality of time and ensemble averages. Instead of the time-average of an observable, f , the central object of study is the time cumulative sum of f :

$$S_N f(x) = \sum_{n=0}^{N-1} f(T^n x). \quad (27)$$

With a finite ergodic invariant probability measure μ , time cumulative sums grow typically linearly with N at a rate which is asymptotically deterministic:

$$S_N f(x) \sim N \int_{\Omega} f d\mu. \quad (28)$$

In infinite ergodic systems, by contrast, time cumulative sums grow sub-linearly with N at a rate which remains random even asymptotically:

$$S_N f(x) \sim a_N \zeta \quad (29)$$

where a_N gives the deterministic rate at which sums accumulate (e.g., $a_N = \sqrt{N}$ or $N^{1-\alpha}$), and the random variable ζ ensures that fluctuations between trajectories never wash out, preserving a genuinely random limit even as $N \rightarrow \infty$. The sub-linear scaling of time cumulative sums explains why ordinary time averages and recurrence frequencies go to zero in such systems. Asymptotic randomness means that different trajectories retain order 1 differences forever. This feature makes infinite ergodic systems fundamentally different from a physical perspective.

Although individual time averages $\frac{1}{N} S_N f(x) \rightarrow 0$ almost everywhere for integrable observables when $\mu(\Omega) = \infty$, an analogue of Birkhoff's Ergodic Theorem persists for relative occupation times. The Hopf Ratio Ergodic Theorem [1] asserts that for any μ -integrable observable f and non-negative μ -integrable observable g with $\int_{\Omega} g d\mu \neq 0$, the ratio of their cumulative time sums converges μ -almost everywhere to the ratio of their spatial integrals:

$$\lim_{N \rightarrow \infty} \frac{\sum_{n=0}^{N-1} f(T^n x)}{\sum_{n=0}^{N-1} g(T^n x)} = \frac{\int_{\Omega} f d\mu}{\int_{\Omega} g d\mu}$$

Thus, while absolute rates of recurrence vanish, the relative frequency with which a trajectory visits different integrable regions of phase space remains non-random and directly recovers spatial measure ratios.

In stochastic processes, the first example of infinite recurrence times that often comes to mind is the simple random walk on $\mathbb{Z}$. Does the simple random walk provide an illustrative example of infinite ergodicity? This is not immediately obvious because the simple random walk is not stationary. We cannot proceed as in the previous section by taking the invariant measure to be the probability measure induced on path space by the stationary law of the stochastic process. However we do not require the invariant measure to be a probability measure (normalisable). What alternative does this enable? The transition probabilities for the simple random walk are:

$$P(i, j) = \begin{cases} \frac{1}{2} & \text{if } i = j + 1 \\ \frac{1}{2} & \text{if } i = j - 1 \\ 0 & \text{otherwise.} \end{cases} \quad (30)$$

The so-called counting measure on $\mathbb{Z}$ is defined as follows. For any $A \subset \mathbb{Z}$:

$$\mu(A) = \begin{cases} |A| & \text{if } A \text{ is finite} \\ \infty & \text{otherwise} \end{cases} \quad (31)$$

In particular, $\mu(\{i\}) = 1$ for all $i \in \mathbb{Z}$. Note that although the counting measure on $\mathbb{Z}$ is not normalisable, it is invariant with respect to the transition probabilities (29):

$$\begin{aligned} \sum_{j \in \mathbb{Z}} P(i, j) \mu(\{j\}) &= \frac{1}{2} \mu(\{i-1\}) + \frac{1}{2} \mu(\{i+1\}) \\ &= \frac{1}{2} + \frac{1}{2} = \mu(\{i\}). \end{aligned}$$

Physically, this simply means that at each step, there is as much probability mass entering each site $i \in \mathbb{Z}$ from neighbouring sites as there is leaving due to hopping. The counting measure on $\mathbb{Z}$ can now be used to induce an invariant (but non-normalisable) measure on the path space of the simple random walk. A slightly unintuitive aspect of this construction is that this induced measure on path space is stationary even though the original stochastic process is not. By taking $f(x) = 1_{\{0\}}(x)$ in (26), we can count the number of visits to site 0 after time N . From the theory of random walks one finds

$$S_N f \sim c \sqrt{N} \quad (32)$$

where c has a non-trivial distribution. Thus ordinary frequencies vanish but normalised occupation counts remain meaningful.

To avoid giving the impression that infinite ergodicity is a complicated way of describing properties of random walks that are obtainable by other means, it is worth briefly considering a second example: the Pomeau-Manneville map. This is a deterministic dynamical system with a compact phase space, $[0, 1]$ where infinite invariant measure arises in a very different way to the simple random walk. The map is

$$T(x) = x + x^{1+\alpha} \mod 1 \quad (33)$$

where $x \in [0, 1]$ and $\alpha > 0$. The important feature is that the fixed point at 0 is neutrally stable because $T'(0) = 1$. Consequently, trajectories that wander close to zero can get trapped for very long periods of time. How long depends on the parameter α . One can show that an invariant measure exists and that near zero it behaves like

$$\rho(x) \sim x^{-\alpha}, \quad (34)$$

which reflects the fact that trajectories have a tendency to accumulate in the vicinity of the “sticky” neutral fixed point. If we consider a set which does not include the neutral fixed point at 0, for example $A = [\frac{1}{4}, \frac{3}{4}]$, it is possible to show that the distribution of the return time, τ_A , has power law tails:

$$\mathbb{P}(\tau_A > n) \sim n^{-\frac{1}{\alpha}}. \quad (35)$$

These tails are the result of some trajectories getting trapped near the fixed point and taking a long time to escape. The parameter value $\alpha = 1$ is special. If $\alpha < 1$ the return time distribution is still integrable, the invariant measure is finite and ordinary ergodic theory applies. If $\alpha \geq 1$, the mean return time diverges and we enter the infinite ergodic theory regime. Note from (33) that the invariant measure becomes infinite since $\int_0^\epsilon \rho(x) dx$ diverges for any $\epsilon > 0$. In this example, infinite measure arises due to the loss of an integrable singularity at a single point in phase space rather

than due to trajectories spreading out over infinitely large regions in the case of the random walk example. Nevertheless many of the same phenomena are observed as we noted for the random walk example. Cumulative occupation counts again scale sub-linearly and after appropriate normalisation converge to a nontrivial random limit. This is the aspect that generalises across many examples: infinite ergodic theory is fundamentally about how power law recurrence time statistics lead to persistent randomness in a broad class of deterministic and random dynamical systems.

For further discussion of infinite ergodicity and a more detailed example in this volume, see [13]. This paper contains explicit calculations for a class of stochastic processes which generalises Geometric Brownian motion to include power law drift and diffusion terms. It is shown how appropriate normalisation can be used to extract physically useful information about the long time asymptotics of such processes even when the corresponding Fokker-Planck equation has no normalisable solution.

5 Non-stationary stochastic processes

Most non-stationary processes are not recurrent, so the notion of infinite ergodicity introduced in Section 4 does not apply to them. It is arguably misleading to refer to such non-stationary processes as “non-ergodic” since ergodicity as described in Sections 2 and 3 is a property of stationary systems, and non-recurrence precludes the more general notion of infinite ergodicity. Nevertheless in some applications, authors do emphasise the non-ergodicity of some “obviously non-ergodic” non-stationary dynamics. This can be because the question of whether individual realisations are representative of the ensemble remains important. It can also be because it is possible to find a transformation which isolates a useful ergodic component of the dynamics, either exactly or approximately.

An illustrative example is provided by the multiplicative random walk [21]:

$$X_{n+1} = z_n X_n \quad (36)$$

with $X_0 > 0$ and the z_i being i.i.d. non-negative random variables with finite mean and variance (the case $z_i \sim \text{Bernoulli}[\frac{3}{5}, \frac{3}{2}]$ is the multiplicative coin-toss example.) We have

$$X_n = X_0 \prod_{i=0}^{n-1} z_i \quad (37)$$

which is clearly non-stationary and therefore non-ergodic. Let us take the logarithm and divide by n :

$$\frac{\log X_n - \log X_0}{n} = \frac{1}{n} \sum_{i=0}^{n-1} \log z_i. \quad (38)$$

Taking the logarithm transforms a multiplicative random walk into an additive random walk. If we now take the limit of large n , the law of large numbers gives

$$\lim_{n \rightarrow \infty} \frac{1}{n} (\log X_n - \log X_0) = \mathbb{E} [\log z] \quad (39)$$

almost surely. Therefore, as n gets large, a typical realisation of (35) almost surely behaves like

$$X_n \sim X_0 e^{\lambda n} \quad (40)$$

where $\lambda = \mathbb{E} [\log z]$ is often referred to as the Lyapunov exponent (since the typical difference between two different initial conditions $X_0^{(1)}$ and $X_0^{(2)}$ grows as $|X_0^{(1)} - X_0^{(2)}| e^{\lambda n}$. On the other hand, since the z_i are i.i.d., the expected value of X_n is given by

$$\mathbb{E} [X_n] = X_0 (\mathbb{E} [z])^n. \quad (41)$$

Therefore as n gets large

$$\mathbb{E} [X_n] \sim X_0 e^{\mu n}, \quad (42)$$

where $\mu = \log \mathbb{E} [z]$. Since the logarithm is concave, Jensen's inequality tells us that $\lambda < \mu$. Typical trajectories are therefore not representative of the ensemble in the sense that the expected value grows exponentially faster with n than individual trajectories. The reason is that the expected value is dominated by large fluctuations which become exponentially rare as n grows but make exponentially large contributions to the average. This mechanism is not a peculiarity of the multiplicative random walk. It appears in many areas of statistical physics such as localisation phenomena, glass dynamics, turbulence and random matrix products. The multiplicative random walk is just the simplest exactly solvable model of a phenomenon that has wide applicability.

Returning to the discussion of ergodicity, the difference between individual trajectories and the ensemble averaged trajectory of the multiplicative random walk is not directly interpretable as a difference of time and ensemble averages. The time average of X_n is either zero or infinite (depending on the sign of λ). The ensemble average is n -dependent and also becomes either zero or infinite as $n \rightarrow \infty$ (depending on the sign of μ). However, there is an underlying Birkhoff-style Equality of Averages ergodicity statement encoded in (38). To see this, define a new variable,

$$\lambda_n = \log X_{n+1} - \log X_n, \quad (43)$$

corresponding to the increment of $\log X_n$, or equivalently the local-in-time growth rate of X_n . From (35) and the fact that the z_i are i.i.d., we note that $\mathbb{E} [\lambda_n]$ is independent of n and given by $\mathbb{E} [\lambda_n] = \mathbb{E} [\log z]$. The left-hand side of (38) is a telescopic sum: $\log X_n - \log X_0 =$

$\log X_n - \log X_{n-1} + \log X_{n-1} - \dots + \log X_1 - \log X_0$. Putting this together, (38) tells us that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \lambda_i = \mathbb{E} [\lambda]. \quad (44)$$

This is Equality of Averages, but for the growth rate of X_n rather than for X_n itself. Note what has happened here: although X_n is trivially non-ergodic, by identifying a transformation of X_n which does possess the ergodicity property as per (43), we are able to understand the temporal behaviour of typical realisations of X_n as per (39).

Empirical analogues of this theoretical procedure abound in practical time series analysis and signal processing. Transformations are commonly used to attempt to isolate a component of the dynamics that is stationary or ergodic (or approximately so in empirical contexts) for observables of interest. The objective is usually to find transformations which can be used to improve some a priori measure of predictive power. In applications, detrending and/or seasonal adjustment may be applied to remove systematic drift or periodic oscillation, differencing may be applied to remove stochastic drift and data transformations may be used to stabilise variance over time. A very commonly used transformation for variance stabilisation is the Box-Cox transformation (which includes the log transformation discussed above as a special case):

$$f_\eta(x) = \begin{cases} \frac{x^{1-\eta}-1}{1-\eta} & \text{if } \eta \neq 1 \\ \log x & \text{if } \eta = 1 \end{cases} \quad (45)$$

and its generalisations. Detrending, seasonal adjustment, differencing and data transformation are often combined to find quantities that best achieve this end. This is reflected in the exactly solvable example of the multiplicative random walk where differencing of the log of the signal is required to obtain the quantity, λ_n , that characterises the typical behaviour of individual realisations of the process. This approach has been very extensively developed in the signal processing and time series analysis literature. The emphasis in these fields tends to be on finding operations which act on non-stationary processes to produce stationary ones, but ergodicity is often implicit in the latter. See for example [10]. For a more complicated example in the present collection, see [23] which exhibits appropriate transformations for processes combining additive and multiplicative increments. A more empirical approach is described in [9] which discusses how to find appropriate transformations from data.

6 Operational notions of ergodicity in practical applications

Establishment of strict ergodicity requires answering a question about the global invariant structure of an entire dynamical system (Ω, μ, T) : do all invariant sets have measure 0

or 1? In applications, researchers often ask a simpler question: are empirical estimators computed from a single trajectory representative of the ensemble of possible trajectories? As we have discussed, these two questions are closely related theoretically. However, it is worth noting that strict ergodicity is neither a strict requirement nor practically necessary for usual applications in physics, economics or psychology. On the one hand, it cannot constitute a requirement since ergodic theorems such as Birkhoff's involve infinite time averages which do not readily translate into real-life applications, as the latter necessarily involve finite-time sequences. On the other hand, it need not be necessary since usual applications in physics and other quantitative fields do not investigate properties of arbitrary observables, but have rather identified a set of pertinent observables to investigate for the model under study. This fundamentally connects with what we discussed in the previous section whereby only specific observables, like the one introduced in (42), display equality of time and ensemble averages for a multiplicative random walk. These shortcomings of strict ergodicity regarding applications have led authors like Khinchin to propose a weaker notion based on typicality that only needs to be met by certain observables. This particular take on the issue is explored in [17]. Furthermore, while for most realistic systems it is close to impossible to answer whether they are strictly ergodic or not, it is often possible to answer approximately whether *some* empirical estimators computed from a single trajectory are representative of the ensemble of possible trajectories. Indeed, in practice, only a finite number of trajectories are observed and each is only of finite length; furthermore, only a limited number of observables are measured and the true underlying dynamics may not even be known. Under such circumstances it is hopeless to attempt to establish indecomposability of an invariant measure. However, in practice, such detailed structural information is rarely required. It may be sufficient that the empirical distributions of finite time and ensemble samples of a particular observable, f , are identical to a given degree of statistical significance. This is referred to in [29] as Equality of Distributions for the observable, f . From the Equality of Distributions perspective, ergodicity becomes a much more operational concept as compared to the metric indecomposability idea. It remains firmly rooted in the fundamental issue of distinguishing between properties of an ensemble and properties of individual members of that ensemble. However, it is no longer a binary property and it is much more broadly applicable and amenable to statistical testing and empirical analysis. Less restrictive still, empirical Equality of Averages for the observable, f , may suffice. Metric Indecomposability, Equality of Distributions, and Equality of Averages form a hierarchy of notions of ergodicity each weaker than the preceding one. For many of the applications in the present collection, Equality of Averages for specific ob-

servables is sufficient to support the arguments being made. This operational approach is quite far from theoretical studies of ergodicity in the mathematical and physical sciences. However, in the preceding sections we have hopefully succeeded in drawing a line from the former to the latter illustrating how these ideas relate to each other. It is not necessary to become an expert in ergodic theory to understand how to perform a statistical test for Equality of Averages in a particular data set, or to make use of the ideas and tools developed to characterise ergodicity in the formal sciences. Nevertheless, it is useful for researchers in applied domains to be aware of how theoretical ideas have developed and how they relate to each other. Such background knowledge may help avoid overloading terminology and introducing confusion. Likewise, theoretical research should not lose sight of the reasons why ergodicity became an important topic of scientific study in the first place. Fundamentally, the interesting issues are not binary questions about whether particular invariant measures are ergodic or not. They are about obtaining a broad understanding of what kinds of averages are reasonable for particular problems and what underlying assumptions are made when we use them.

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